

ECEC state demand inputs: method

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1 Purpose

The simulator reads one file for each geography. The file holds six rows, one for each care type, and each row holds the annual hours of child care. This memo gives the source of those numbers.

The build covers five geographies: Kentucky (KY), New Jersey (NJ), New York (NY), New York City (NYC) and Virginia (VA). Every source is linked where it is first used.

Why the build exists

The simulator must know the mix of child care in each state, and the survey that measures that mix cannot supply it. [NSECE](#) does not record the state of a family. It reports the Census region and nothing below it, so no filter recovers a state from it.

I therefore build the mix from the administrative records of each state, and I use NSECE only for the care that no administrative record shows. Two rules follow from that, and they hold across every step below.

- The build works at the state level. It reads county data one time, in step 1, and it adds the counties together immediately. No later step uses a county, so the build needs no county crosswalk and no county name matching.
- The build prefers a state measurement to a national one. It uses a national figure only where the state publishes nothing, and it says so each time it does.

The position of NYC

NYC is a geography of its own here, and every step below gives its figures. It is also inside NY, and the NY figures include it. The city appears twice for two reasons. The New York City Department of Health and Mental Hygiene regulates its centers, not the state agency that regulates everywhere else. The city is also asked about on its own.

The two builds do not agree, and that is deliberate. Each one takes its own control total in step 2, and step 5 calibrates each one separately. The five boroughs therefore come out differently in the two builds. Read one or the other. Do not add them together.

2 Results

The build counts two quantities and calculates a third as a remainder. Every row in the output divides one of the three.

Table 1: Children in non-parental care, by broad setting

	Center-based	Not center-based	Total	Share center
KY	83,719	78,211	161,930	52%
NJ	173,570	136,376	309,946	56%
NY	283,125	366,704	649,829	44%
NYC	138,604	163,372	301,976	46%
VA	135,286	127,714	263,000	51%

Table 2: Care that is not center-based, counted and calculated

	Licensed home care	Informal care
KY	2,363	75,848
NJ	1,643	134,734
NY	84,223	282,481
NYC	32,103	131,269
VA	4,525	123,189

NY and NYC hold far more licensed home care than the other three geographies, and the difference is real rather than an error. The New York Office of Children and Family Services registers a home provider that keeps three children or more. Its register therefore holds care that leaves no trace elsewhere. Kentucky, New Jersey and Virginia register only larger providers. That care exists in those states too, but it falls into the remainder instead.

The figures below are counts of children. They are not record counts, and they are not weighted survey figures.

Table 3: Children by care type

	KY	NJ	NY	NYC	VA
Center-Based	39,943	116,747	180,378	92,876	87,410
Unpaid Center-Based	43,776	56,823	102,747	45,728	47,876
Paid Home-Based	2,363	1,643	84,223	32,103	4,525
Other Unpaid	45,830	81,411	170,686	79,318	74,436
Other Paid	27,363	48,607	101,909	47,357	44,442
Unpaid Home-Based	2,655	4,715	9,886	4,594	4,311
Total	161,930	309,946	649,829	301,976	263,000

Sub-step 6.3 gives the hours for each child, which come from [NSECE](#).

Table 4: Annual hours by care type, in millions

	KY	NJ	NY	NYC	VA
Center-Based	67.3	196.8	304.1	156.6	147.3
Unpaid Center-Based	62.9	81.6	147.6	65.7	68.8
Paid Home-Based	4.5	3.2	161.7	61.6	8.7
Other Unpaid	88.4	157.1	329.3	153.0	143.6
Other Paid	42.5	75.5	158.4	73.6	69.1
Unpaid Home-Based	5.6	10.0	21.0	9.7	9.1

3 The build

Step 1. Count the children

I read the [Census population estimates](#) for 2019, keep children of age 0 to 4, and add the counties together for each geography.

Table 5: Children age 0 to 4, 2019

Counties	Children 0–4, 2019
KY	120
NJ	21
NY	62
NYC	5
VA	133

NYC uses five counties from the same file: Bronx 100,199; Kings 182,926; New York 76,144; Queens 137,095; Richmond 27,354. The city holds 46 percent of the children in NY.

The build stops at age 4 for three reasons.

- The simulator stops there. Its ACS processing keeps children below age 5 in seven places, so a wider demand file would anchor the model against households it never sees.
- A universe of ages 0 to 5 raises each control total by 15 to 23 percent. That rise combines a population about 19 percent larger with a survey rate that moves, because five-year-olds use non-parental care less than younger children do. Most of them are in kindergarten, which the survey does not count as child care.
- Census publishes no county estimate for age 5 on its own. The build can count a state at that age, but it could only estimate NYC, so the city would stop being counted and start being modeled.

Step 2. Find the children in non-parental care

I read the [NSCH household survey](#) for 2019. I keep the children of age 0 to 4, weight each child by the survey child weight, and take the share in any non-parental care. That share times the population from step 1 is the control total. Every later count is a share of it.

Table 6: The non-parental care rate and the control total

	Children 0–4	Rate	In non-parental care	Records
KY	272,610	59.4%	161,930	147
NJ	514,690	60.2%	309,946	101
NY	1,127,001	57.7%	649,829	112
NYC	523,718	57.7%	301,976	—
VA	505,477	52.0%	263,000	121

- The control total is always a 2019 figure, because both of its inputs are 2019 data. This is what allows the build to read a register published in 2026 without ageing it backward: the register decides how care divides across types, and it never decides how much care there is.
- NYC has no records of its own, which is why its row shows a dash. It takes the rate of NY, because NSCH publishes no geography below the state. Every difference between the city and the state therefore has to come from the registers in step 3, and it does.
- This rate is the weakest number in the build. Each geography rests on about 120 families, which is a standard error of five to six percentage points on a figure that multiplies every later count. Sub-step 2.1 explains why the build still uses a single survey year.

2.1 Why the rate uses one survey year

Pooling survey years would cut the sampling error on the non-parental rate by 40 to 47 percent. No other change available here would buy as much accuracy for as little work. The build does not pool, because every input has to sit on the simulator’s 2019 base year.

A pool of 2018, 2019 and 2020 has its midpoint exactly at 2019, so it is the one pool that passes the vintage test. It fails on content instead. NSCH collected its 2020 data through the pandemic, and the use of child care collapsed.

Table 7: Non-parental care rate by survey year

	2018	2019	2020
KY	57.3%	59.4%	45.8%
NJ	58.3%	60.2%	42.6%
NY	64.7%	57.7%	62.1%
VA	61.7%	52.0%	46.7%

All four rows come from [NSCH](#). The fall in 2020 is a real change rather than sampling noise, and pooling it in would pull every total down by several points. A pool of 2018 and 2019 alone avoids that, but its midpoint falls in the middle of 2018 while every other input is 2019.

So the rate keeps its full sampling error. The memo reports that error rather than trading it for a bias that nothing downstream could recover. The repository holds all three survey years, so anyone can repeat the test.

Step 3. Convert the registers into children

Every state licenses child care providers and publishes a register, and every register reports capacity rather than children. Three sub-steps convert one into the other.

3.1 Remove the school-age capacity

Each register mixes school-age capacity into its figures, and each one hides it differently.

- Kentucky’s Division of Regulated Child Care publishes no bulk file, so the build reads [Child Care Aware of America’s republication](#) of its licensed and certified registers. Neither register says which providers are centers and which are homes, so the build applies the state licensing rule: a licensed home keeps 12 children or fewer. That gives 1,609 centers and 335 homes. The build drops a provider that serves school-age children only. A mixed age band scales by a ratio between 0.487 and 0.827. The build measures each ratio inside the register itself. It compares the mean capacity of a band against the same band without school-age children.
- The New Jersey Department of Children and Families (DCF) publishes [its register of licensed centers](#), with an age range for each one. The build drops the 1,035 centers serving ages 6 to 13 only, and scales a mixed range by 0.820. That scale is also measured inside the register, from two band comparisons that give 0.8224 and 0.8183.
- The New York Office of Children and Family Services (OCFS) publishes [its register](#) with capacity broken out by age, which is the clean case. Infant plus toddler plus preschool gives the under-five capacity exactly, so the build applies no scale. Homes report a single total, but the register publishes their school-age capacity separately and the build subtracts it.
- NYC draws on both agencies. Its centers come from the New York City Department of Health and Mental Hygiene (DOHMH), whose [permit file](#) gives one capacity figure per permit and covers children below age 6 by construction. Its homes come from the OCFS register and take the same subtraction as NY.
- The Virginia Department of Social Services (VDSS) writes the age range as text in [its register](#). The build parses it into months. It drops any provider starting at age 5 or later, and scales a center reaching past age 6 by 0.60. VDSS appends a contact block to that field, and the build strips it before parsing. Sub-step 3.4 gives what that cost when it was missed. Read correctly, a span ending below age six holds 0.527 of one running to nearly thirteen in Child Day Centers. It holds 0.604 in Religious Exempt centers. The two pool to 0.540, so 0.60 is the conservative end of the measured range. Homes take no scale at all, because VA caps a family day home at twelve children whatever ages it serves.

3.2 Move the register back to 2019

Three of the registers are 2026 exports and have to be brought back to the base year.

- NY scales each provider type separately, against the OCFS series of provider counts and capacity by modality covering 2015 to 2025. Group family day care scales to 0.86 of its 2026 size, and OCFS centers to 0.93. The DOHMH centers that NY also carries scale at 1.00, as they do in the NYC build. Homes have closed in numbers that centers have not, so one blanket factor would distort the mix rather than correct it.
- NYC takes its own factors from the same OCFS series, which publishes the city separately from the rest of the state. Group family day care scales to 0.84 in the city. City centers scale to 1.00, because the city register held almost the same capacity in 2019 as it does now.

- VA scales by one statewide factor of 0.912. The build computes it from VDSS’s [Child Care Data Matrix](#) for June 2020, as that report’s child care slots divided by the current register’s capacity. The denominator is cut to the six approval routes the 2020 column names in its own header, so that it counts the same universe as the numerator. Dividing by all ten routes put 4,597 places into the denominator that the 2020 figure never counted, and gave 0.901 instead.
- KY and NJ need no correction, because their registers already sit at the base year. The Child Care Aware republication is a September 2018 snapshot, and the DCF register is frozen at May 2019.

3.3 Convert capacity into children

Capacity is a legal maximum, so real enrollment is lower and something has to bring it down.

- KY needs no fill rate, because it measured its own enrollment. The [2019 Kentucky Workforce Study](#) prints capacity and enrollment side by side in Table 20, giving 42 children for each center and 9 for each home. This is the preferred method, and it is why 1,609 centers yield only 67,578 children.
- NJ measured its own fill rate. The [2021 NJ Child Care Market Rate Survey](#) puts its centers at 67 percent full in its 2020 column.
- NY, NYC and VA have published no fill rate, so they borrow the national figures: 90 percent for a center, from the ACF report on [center-based providers](#), and 60 percent for a home, from the ACF report on [home-based providers](#). Sub-step 3.5 gives the evidence that both are too high.
- NY and NYC homes use 80 percent instead of 60. The 60 percent figure is itself a product of a fill rate and an under-five share, and the OCFS register already removed the school-age capacity in sub-step 3.1. Applying the full figure would remove school-age children a second time, so those homes take the fill rate alone.

Table 8: Providers, capacity and children, from the registers

		Providers	Under-5 capacity	Children
KY	Centers	1,609	107,369	67,578
KY	Homes	335	2,115	3,015
NJ	Centers	3,128	251,159	169,106
NY	Centers, state	2,073	153,053	138,191
NY	Centers, city	2,310	141,300	127,580
NY	Homes	11,918	117,401	93,381
NYC	Centers	2,310	141,300	127,580
NYC	Homes	4,580	46,057	36,634
VA	Centers	3,186	190,975	172,432
VA	Homes	1,935	12,792	7,700

Providers and capacity come from the five registers named in sub-step 3.1: Child Care Aware for KY, DCF for NJ, OCFS for NY, DOHMH for NYC centers, and VDSS for VA. Children come from the conversion in sub-step 3.3.

NY appears twice in that table because two agencies regulate child care there. OCFS licenses providers across all 62 counties, but it licenses no center inside New York City, because DOHMH permits those itself under Article 47 of the city Health Code. The OCFS register therefore shows zero centers in the five boroughs, and that is the correct result rather than a failed match. OCFS does license home providers inside the city, so NYC reads centers from DOHMH and homes from OCFS.

DCF licenses centers only, so NJ has no register of home providers at all. Its home care comes from the same market rate survey as its fill rate, which counted 1,396 children below age 5 across 913 responding providers. The build raises that to the 1,254 registered providers in the state, giving 1,917 children before calibration. Every count in Results is after calibration; every count in the table above is before it.

3.4 A parse defect in the Virginia register

VDSS writes the age range and a contact block into one field:

```
1 month - 12 years 11 months VDSS Contact:
Tara K Martin: (804) 588-2312
```

The build reads the upper bound as the text after the last hyphen. In a string like that one, the last hyphen falls inside the telephone number. The upper bound therefore became 2312. That text holds neither “year” nor “month”, so it parsed to zero months.

A provider whose upper bound reads zero looks like a provider that serves no school-age children, so the build kept 100 percent of its capacity. The rows affected mostly run from one month to twelve years and eleven months, which is the most mixed span the register holds.

Table 9: Rows affected by the Virginia parse defect

	Affected
Rows	1,055
Licensed places	78,486
Share of the VDSS register	21.2%
Religious Exempt Child Day Centers	861
Voluntarily registered family day homes	173
Certified pre-schools	21

Stripping the contact block before parsing resolves every one of them, and it changes Virginia alone. The table below isolates that one change; two later corrections moved the factor again – the rebase denominator in sub-step 3.2, and the program overlap in sub-step 4.1 – to 0.588.

Table 10: Virginia before and after the parse fix

	Before	After
Mixed-span centers scaled	2,965	3,591
Center capacity below age 5	207,835	188,600
Center children	187,654	170,287
Calibration factor	0.62	0.71

Two things follow from this, and both matter more than the numbers.

The calibration concealed the defect. Step 5 forces Virginia’s formal care to the regional share whatever the register says. The extra 19,235 places therefore went into a more extreme scale factor, not into an obviously wrong total. What looked like Virginia having an unusually complete register was partly a parse failure.

The scale itself survived the fix. The 0.60 share was first measured on the uncleaned strings and happened to land in the right place. Measured on the cleaned field it is 0.527 for Child Day Centers and 0.604 for Religious Exempt centers, so 0.60 remains defensible. The value did not change. What changed is that it now rests on a field the build reads correctly.

The build now stops if any Virginia age range parses to an upper bound of zero months. The same failure cannot return silently if VDSS changes the field.

3.5 Why the borrowed fill rates are too high

Two states have published their own figures, and both point the same way.

KY measured its centers at 57 percent full in 2019, against the 90 percent that NY, NYC and VA borrow. VA measured a related quantity in 2022, asking 1,968 providers what they can actually serve and comparing the answer to their license: centers reported 73 percent of their authorized capacity and homes 91 percent, with 121,260 licensed places in that sample against 89,374 children the providers said they could take. That study is [EdWorkingPaper 24-983](#).

The build uses neither figure as a rate. KY’s applies to KY. VA’s is from 2022 and measures capacity rather than enrollment, so it would duplicate a correction the fill rate already makes. What both establish is the direction and rough size of the error: licensed capacity overstates real supply by a quarter to a third, and the national fill rate does not remove enough of it.

Step 4. Add the programs that count their children

Head Start, Early Head Start and public pre-K publish actual enrollment, so the build takes the published count rather than estimating one.

Table 11: Program enrollment, as published

	Pre-K	Head Start	Early Head Start	Total
KY	28,465	12,307	3,004	43,776
NJ	41,305	11,940	3,578	56,823
NY	54,451	36,863	11,433	102,747
NYC	26,717	13,073	5,938	45,728
VA	33,790	11,579	2,507	47,876

- Pre-K comes from the school-level records of the NCES [Common Core of Data](#) for 2018–19, counted school by school.
- Head Start comes from the NIEER [State of Preschool 2019](#), and Early Head Start from the NIEER [report on Head Start](#). Both are labelled ages 3 and 4, and birth to three, so every figure here sits inside the 0 to 4 universe. The two NIEER publications disagree slightly on Head Start in Kentucky: the yearbook gives 12,307 and the Head Start report 12,163. The build uses the yearbook figure for consistency with the other four geographies.

- NYC takes its Head Start figure as a share of NY, using the ACF [service location file](#): 7,194 funded places sit in the five boroughs against 20,285 in the state, so the city takes 35 percent of the state level.
- The build subtracts these children from the licensed center count rather than adding them, because a program center that holds a state license was already counted in step 3. It subtracts only the share that really is in the register, which sub-step 4.1 measures.
- A published count does not move in step 5. The build treats it as a census, so the calibration falls entirely on the capacity-derived counts.
- This row is an upper limit everywhere. Head Start and state pre-K overlap in an unknown number of children, and no source publishes the size of that overlap.

4.1 How much of a program is already in the register

A program site that holds a state license sits inside the capacity step 3 converted, so its children would be counted twice unless they are subtracted. A site the register never lists does not, and subtracting it removes children who were never counted in the first place. Until 18 September 2026 the build subtracted everything, which assumed the first case always holds.

I measured it, for all three programs. I matched every open Head Start service location in the federal file to the register. I did the same for every public school that reports pre-K enrollment. The match uses ZIP plus street number plus street name. It also uses coordinates within 25 metres, where both sides carry them. Head Start is weighted by funded places, and pre-K by enrolled children. School addresses come from [NCES EDGE](#) and pre-K counts from CCD school by school; those counts sum to exactly the figures in step 4, which confirms where those constants came from.

Table 12: Share of program children at an address the register also holds

	Pre-K	Head Start	Early Head Start	County build
KY	37.0%	37.1%	50.6%	38.1%
NJ	55.4%	64.7%	62.7%	63.0%
NY	52.2%	77.3%	77.5%	49.3%
NYC	31.2%	70.6%	67.9%	—
VA	register publishes no address; takes the mean			—

The last column is the same quantity measured independently by the earlier county build, for Head Start only. KY and NJ agree to within 1.7 percentage points, which is the check on this method. NY differs because the match here reads both regulators, OCFS and DOHMH, and much of the state’s Head Start sits in the city.

Pre-K overlaps less than Head Start everywhere, which is what the licensing rules predict: a pre-K classroom inside a school building is usually exempt, while a Head Start center commonly holds a license. NYC is the lowest at 31 percent, and it runs the largest school-based pre-K program of the five.

Three caveats.

- A shared address is not proof that the same children were counted twice. A licensed provider at a school address can be a before-school or after-school program rather than the pre-K

classroom. The match sets an upper limit on the duplication. It does not confirm it child by child.

- The federal files are 2026 exports and the registers are not, so a site that has since closed or moved fails to match. Every figure is therefore a lower limit, and KY's is the least reliable because its register is from 2018.
- VA cannot be matched at all, so it takes the mean of the other four. That is the least defensible number in the table, and VA is also the geography the calibration moves furthest.

Step 5. Calibrate the total to the survey

A register describes the relative distribution across care types well and the total badly: it puts far more children in formal care than any household survey finds. The build corrects the total with one factor and leaves the distribution alone.

1. Take the share of non-parental care that is formal. [NSECE](#) gives it by Census region: 56.5 percent in the Northeast and 53.2 percent in the South.
2. Multiply that share by the control total from step 2 to get the target.
3. Hold the program counts from step 4 fixed.
4. Scale every other count by one factor until formal care reaches the target.

Table 13: The calibration factor and the resulting split

	Scale factor	Formal	Informal	Informal share
KY	0.78	86,082	75,848	47%
NJ	0.86	175,212	134,734	44%
NY	0.90	367,348	282,481	44%
NYC	0.88	170,707	131,269	44%
VA	0.59	139,811	123,189	47%

The build computes each factor from the two totals, so nothing here is chosen by hand. This table is the primary check on everything above it, and it reads as follows.

- Every factor is below one, which is the direction the evidence predicts. Capacity is a ceiling and the borrowed fill rates are too high, so a register should always overstate formal care.
- NJ, NY and NYC sit between 0.86 and 0.90, close together despite three different agencies keeping their registers – DCF, OCFS and DOHMH – and despite two of the three borrowing their fill rates while the third measured its own.
- KY falls least, to 0.78, and it is the only state that measured how full its own centers are: 57 percent against the 90 percent that NY and VA borrow. A state using its own fill rate needs less correction than one that borrowed a figure known to be too high, which is the ordering here.

- VA falls furthest, to 0.59. The VDSS register is the most complete of the four, listing unlicensed and voluntarily registered family homes that no other state here records, and it also borrows the national fill rate. Both raise its count of formal care.
- The spread has narrowed twice as assumptions were replaced by measurements: 0.62 to 1.58 when every program child was subtracted, 0.67 to 1.17 once Head Start was measured, and 0.59 to 0.90 now that pre-K is measured too. An earlier run put three factors near 1.00, which looked reassuring but was an artifact of subtracting too much.

Step 6. Complete the rows

6.1 Use no price split

An earlier version of this build divided center care into low-priced and high-priced. Nothing in the state data supports that division. A state publishes one price. Every state therefore fell wholly on one side of the national boundary. A care type left with no children then looks unmeasured rather than genuinely absent.

Center care is now a single row. It carries 1,686 hours for each child each year, which is the child-weighted mean across the two price bands that [NSECE](#) distinguishes nationally.

6.2 Calculate the care that no register records

Three of the six types appear in no register anywhere: unpaid relatives, nannies and informal arrangements. Unpaid relative care is not a business, files no return, and leaves no administrative trace of any kind.

The build subtracts all counted care from the control total. It treats the remainder as this care, and divides that remainder three ways on the national pattern from [NSECE](#). The remainder runs 44 to 47 percent of non-parental care across the five geographies, against 47 percent nationally in the same survey.

6.3 Convert children into hours

Each care type carries a measured hours figure from [NSECE](#): the child-weighted mean weekly hours reported for the child's primary arrangement, multiplied by 50 weeks. The figures run from about 1,440 hours a year for program center care to about 2,120 for unpaid home care. The hours column is therefore not proportional to the children column.

4 Limits

1. The build cannot see unpaid relative care, which is the largest informal category nationally. That figure is a remainder rather than a measurement, and no administrative source would fix it. Only a restricted-access survey file with a state identifier would.
2. Three of the six rows carry no state information. Other Paid, Other Unpaid and Unpaid Home-Based are a national ratio applied to a residual, and they are 44 to 47 percent of all care in the file. The state content lives in the other three rows, and really in the population, the survey rate and the ratio of center to home capacity.
3. Paid Home-Based is not comparable across states. NY shows 84,223 children and VA 4,525, and that gap is mostly licensing law rather than behavior: OCFS registers a home provider

keeping three or more children, and the other states register only larger ones. NJ has no home register at all.

4. The mix carries the date of the register. The total is 2019 by construction, but how it divides across care types reflects whenever the state last published. Composition genuinely drifts, so a 2026 register shows too few family homes for 2019 even after the vintage correction in sub-step 3.2.
5. Unpaid Center-Based is an upper limit everywhere, because Head Start and state pre-K overlap by an unpublished amount and a program site may also hold a state license.
6. Nothing propagates the uncertainty. The survey rate carries a standard error of five to six percentage points, which is roughly ten percent on every number in every file, and the output is bare point estimates.
7. Step 5 forces an exact match to the survey. That is a decision rather than a finding. The alternative is to let the registers stand and report the disagreement instead of removing it.
8. The simulator still expects seven rows, with center care split by price, and it rejects a file with six. Bridging the two is one mechanical operation that divides the center row on a national share. The build keeps that operation outside the method, so that it never appears to measure a quantity it cannot measure.